



A. Course Objectives and Outcomes

Course Objective and Outcome Form

Department of Electrical and Computer Engineering

School of Engineering and Physical Sciences

North South University, Bashundhara, Dhaka-1229, Bangladesh

1. Course Number and Title

EEE111 / ETE111 – Analog Electronics I

2. Number of Credits

3 Credits (Theory) + 1 Credit (Lab)

3. Type

Required, Engineering, Lecture and Laboratory

4. Prerequisites

EEE141 – Electrical Circuits I

5. Contact Hours

Lecture – 36 hours

Laboratory – 36 hours

6. Course Summary

This course introduces the characteristics, operation, and applications of fundamental semiconductor devices such as diodes, bipolar junction transistors (BJTs), and field-effect transistors (FETs). The course emphasizes analytical techniques for analog electronic circuits using appropriate mathematical models and graphical interpretation. Practical circuit configurations used in real-world applications are introduced, and device behavior is explained using equations, characteristics curves, and laboratory experimentation.

7. Course Objectives

- To introduce the operating principles and characteristics of semiconductor devices.
- To explain the analysis techniques of analog electronic circuits.
- To develop understanding of transistor biasing and amplifier configurations.
- To develop practical skills in constructing, testing, and troubleshooting analog circuits.

8. Course Outcomes (COs)

CO	Course Outcome Description	Weightage (%)
CO1	Explain the characteristics and operation of diodes, BJTs, and FETs.	25
CO2	Analyze simple electronic circuits using diodes and transistors.	25
CO3	Apply small-signal models of BJTs and FETs for amplifier analysis.	25
CO4	Conduct experiments and analyze, interpret, and verify experimental data from analog electronic circuits.	25

9. Mapping of CO–PO

CO	POs	KP	Bloom's Level	Delivery Methods	Assessment Tools
CO1	PO1	K3	Cognitive / Understand	Lectures	Quiz, Exam
CO2	PO1	K3	Cognitive / Analyze	Lectures, Problem Solving	Quiz, Exam
CO3	PO1	K3	Cognitive / Apply	Lectures, Examples	Exam
CO4	PO4	K6	Psychomotor / Manipulation	Laboratory Experiments	Lab Performance, Reports

10. Resources

• Textbooks

1. Boylestad, R. L., & Nashelsky, L. *Electronic Devices and Circuit Theory*, 11th Edition.
2. Sedra, A. S., & Smith, K. C. *Microelectronic Circuits*, 6th Edition.

11. Weekly Course Schedule (Theory)

(13 Weeks, 36 Contact Hours)

Week	Topics Covered	Book Sections	Assessment
1	Semiconductor Materials, PN Junction, Diode Characteristics	1.1–1.11	
2	Zener Diode, LED	1.15–1.16	–
3	Diode Applications: Rectifiers, Clippers, Clampers	2.1–2.11	Quiz 1
4	BJT Construction, Operation, Configurations (CB, CE, CC)	3.1–3.7	–
5	DC Biasing of BJT Circuits	4.1–4.8, 4.16	Quiz 2
6	Midterm Examination	–	Midterm
7	Small Signal AC Analysis of BJT Amplifiers	5.1–5.10	–
8	FET Operation and Characteristics	6.1–6.11	Quiz 3
9	FET Biasing Techniques	7.1–7.8	
10	FET Amplifier Circuits	8.8–8.11	–

11	Review and Problem Solving	–	Quiz 4
12	Advanced Applications	–	–
13	Final Examination	–	Final

12. Weightage Distribution among Assessment Tools

Assessment Tools	Weightage (%)
Attendance	10
Quizzes	20
Midterm Exam	30
Final Exam	40

13. Grading Policy

As per North South University grading policy:

<https://www.northsouth.edu/academic/grading-policy.html>

14. Course Policies

- Regular attendance in both lecture and laboratory sessions is mandatory.
 - Students must complete laboratory experiments and reports individually.
 - No make-up examinations will be conducted without prior approval.
 - Academic integrity must be strictly maintained.
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15. Course Instructor Information

Instructor: Acramul Haque Kabir

Office: SAC 1046B

Email: acramul.kabir@northsouth.edu

16. Class Timing and Office Hours

Office Hours & Class Schedule: [Click here to view](#)